SG Simulation Model Documentation

Information

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Summary:	This document describes functionality of SG Simulation Model and provides a guide to its usage and control.		

History

Version:	Date	Description
0.1	2022-02-25	Initial version
0.2	2022-04-27	Added general SV communication
0.3	2022-05-03	DPS communication extended to all IEDs
0.4	2022-06-30	Changed simulation model base
0.5	2022-09-15	Added SCADA section and performance thresholds
0.6	2022-12-09	Edited SCADA text according to new updates
1.0	2022-12-12	Text updates and release to testing
1.1	2023-03-07	Smart Grid topology extensions – circuit breakers
1.2	2023-03-13	Internal attacker + FDI attack integration
1.3	2023-04-20	DoS attack integration

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Abbreviations

Abbreviation	Definition
ASDU	Application Service Data Unit
DoS	Denial of Service
DPS	Digital Primary Substation
DS	Digital Substation
DSS	Digital Secondary Substation
GOOSE	Generic Object Oriented Substation Events
GUI	Graphical User Interface
GW	Gateway
HMI	Human Machine Interface
IEC	International Electrotechnical Commission
IEC104	IEC 60870-5-104
IED	Intelligent Electronic Device
IOA	Identification Object Address
IPDV	Inter-Packet Delay Variation
OS	Operating System
RAM	Random Access Memory
SCADA	Supervisory Control And Data Acquisition
SV	Sample Value
SW	Switch
VM	Virtual Machine
WAN	Wide Area Network

Introduction

The simulation model is provided in form of the open virtualization format archive (.ova file). The file contains an image of a virtual machine, and it can be imported using virtualization applications such as VMware Workstation or Oracle VM VirtualBox.

The simulation model is using the following components:

- Operating system: Linux Lite 6.0
- Mininet network emulator: 2.3.0d6
- Libraries for communication emulation
  - lib60870: 2.2.0
  - libIEC61850: 1.4.2.1

Technical information

The model network topologies are created in the Mininet network emulator[1]. The emulator allows to build network topologies composed from hosts and switches and because of its use of lightweight virtualization with a shared kernel, the emulator is very resource friendly.

The communication is emulated using open-source libraries[2] with modified scripts.

[1] https://mininet.org/
[2] https://libiec61850.com/

Model topologies

The model contains two topologies – basic and extended. Topologies include:

- Basic topology
  - 2x Digital secondary substations
  - Control center
- Extended topology – extends the basic topology with DPS, so it contains:
  - 2x Digital secondary substations
  - 1x Digital primary substation
  - Control center

The basic topology

The basic topology is shown in Figure 1. All devices' names are used consistently in the model implementation. The model contains two digital secondary substations and a control center. Both substations have a switch inserted between gateway and RTU. These switches are configured for external connections called *AttackDSS1* and *AttackDSS2*. Any other virtual machine can be connected to these connections using the *Internal Network* option with the corresponding name. This option is described in section Connecting an external machine. The topology includes IEC104 communication.

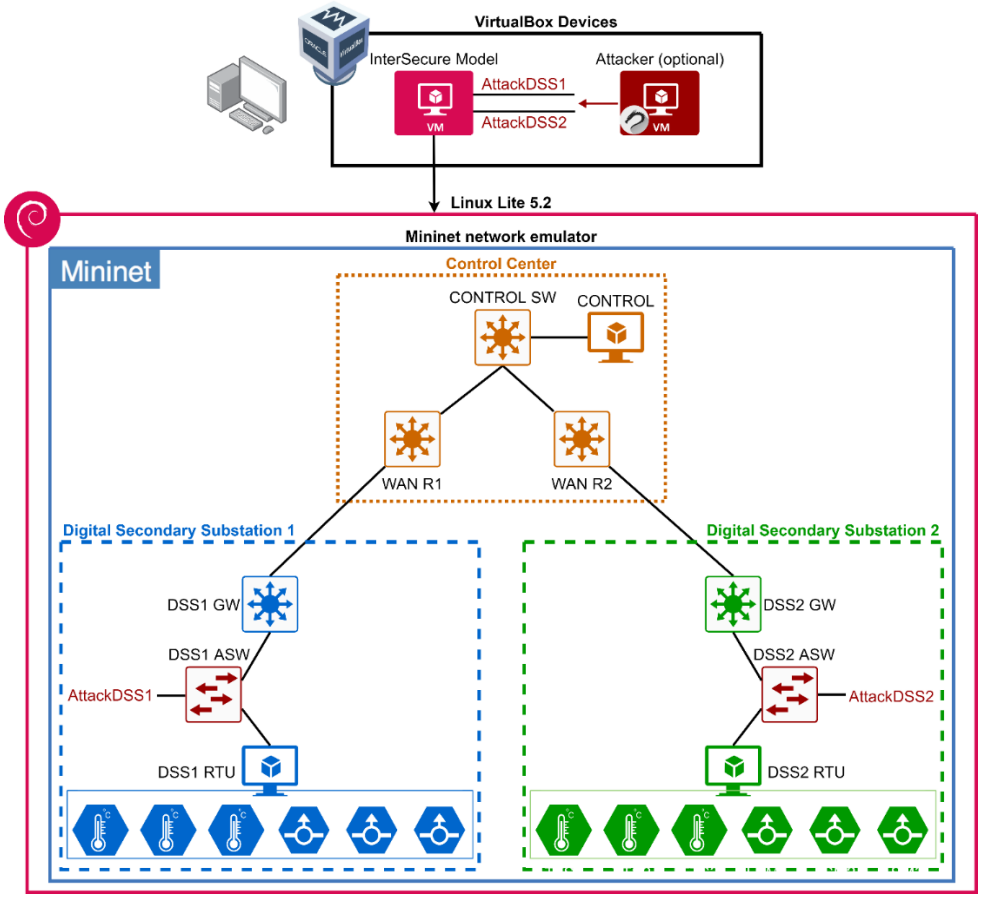

Figure 1 – Basic topology with two digital secondary substations

The extended topology

The extended topology adds a digital primary substation as shown in Figure 2. The substation is connected to the control center, and it contains different device types (IEDs and HMI) and also different types of communication (GOOSE, SV).

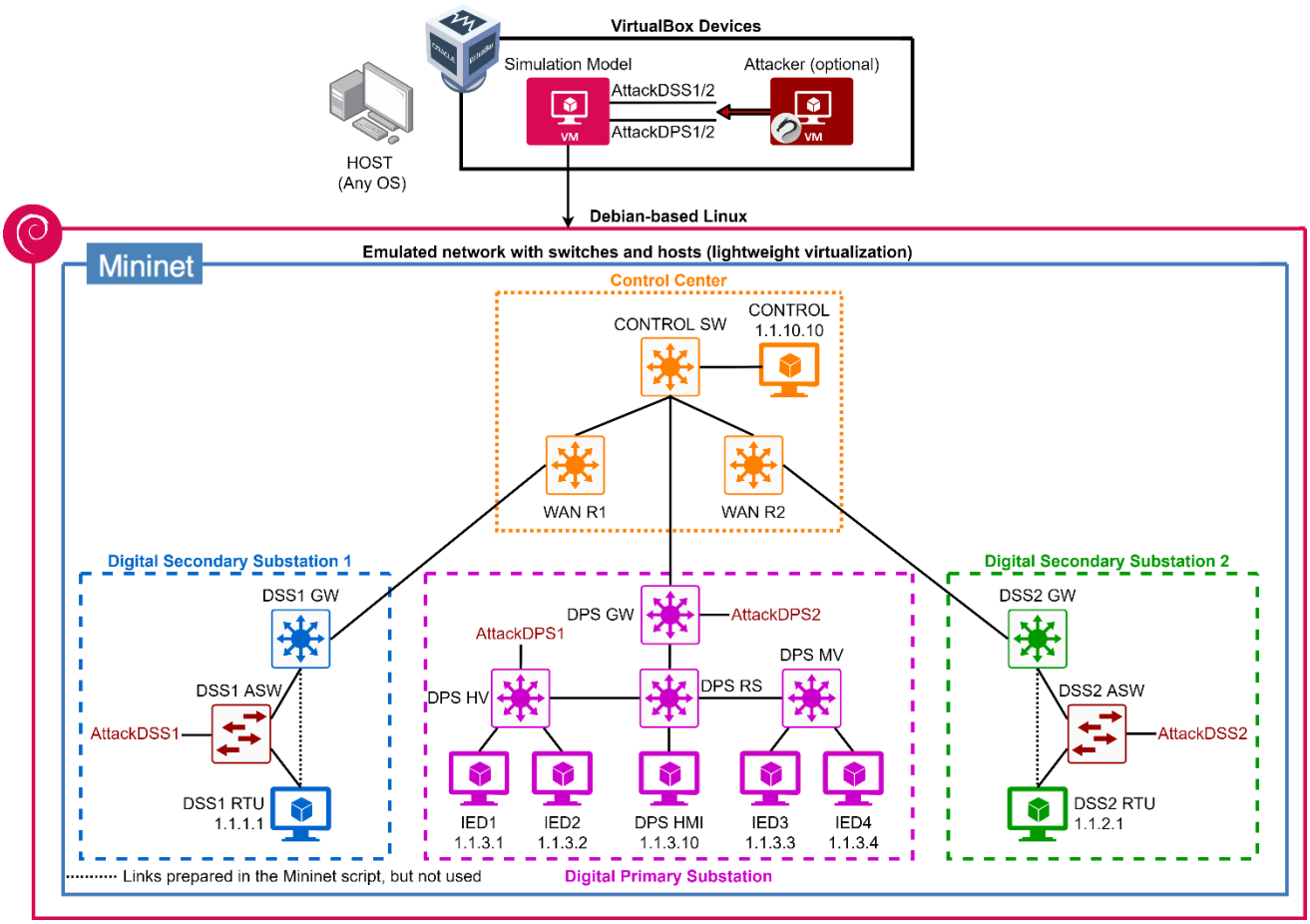

Figure 2 – The extended topology

Moreover, the extended topology includes internal attacker's device and simulation of circuit breakers as shown in Figure 3.

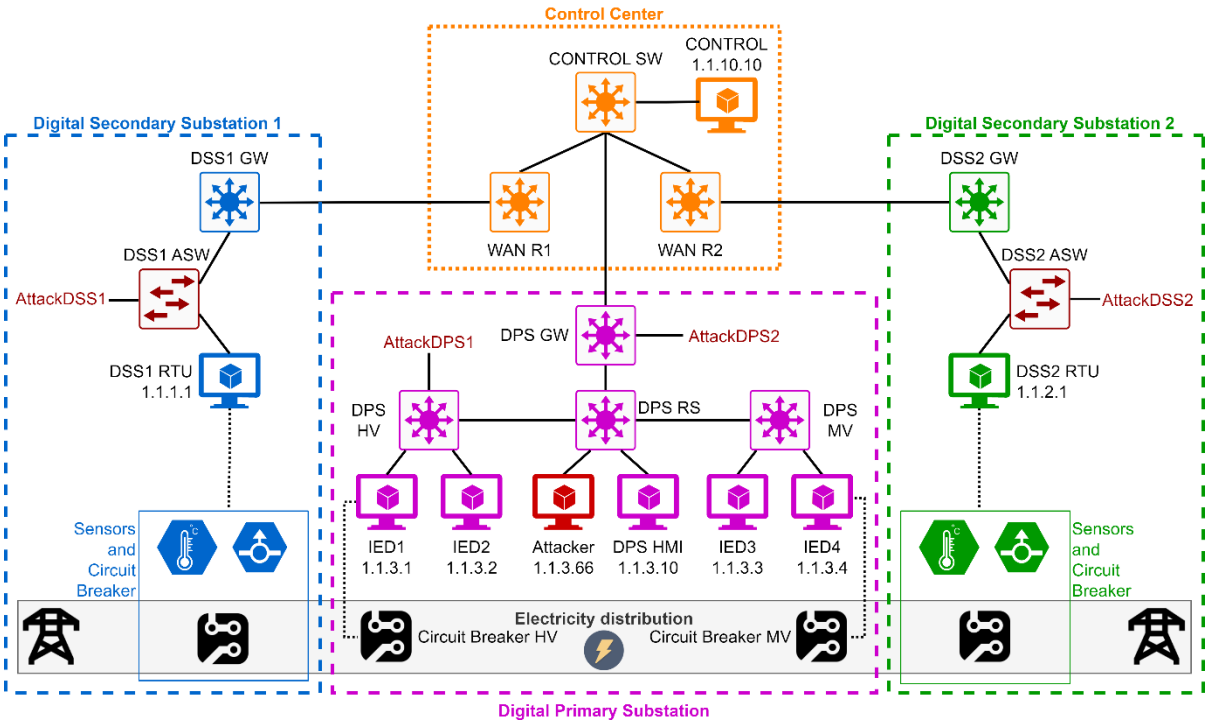

Figure 3 – The extended topology with the internal attacker and circuit breakers

Model communication

IEC 60870-5-104

Emulated communication in the basic simulation model uses IEC 60870-5-104 (IEC104) standard and two modes of communication between RTU and CONTROL devices:

1. Periodical – RTUs are sending *I type* messages every 1 second. These messages contain data from 5 sensors (IOAs) with randomly generated values from defined intervals. Every *I type* message is acknowledged by an *S type* message.
   Generated values are:
   - IOA 15: 52.2 – 52.9
   - IOA 16: 45.5 – 47.9
   - IOA 19: 0.25 – 0.27
   - IOA 23: 45.5 – 47.9
   - IOA 24: 0.003 – 0.105
2. Read request – the control device is sending *I type* request in randomly generated time periods from the interval 1 – 60 seconds. The RTU replies with an *I type* data message, which is then acknowledged by an *S type* message.

IEC 61850 Generic Substation Events (GOOSE)

Emulated communication in a digital primary substation uses GOOSE and SV messages. GOOSE communication uses layer 2 multicast between IEDs (IED1 and IED4) and the HMI. GOOSE messages are blocked on the DPS GW and are therefore not leaving the primary substation.

IEC 61850 Sampled Values (SV)

SV messages uses subscriber – publisher model. They are being sent between IEDs (IED2 and IED3) and the HMI and similarly to GOOSE, they are blocked on the DPS GW and therefore do not leave the primary substation. They do not use VLAN tagging.

Import

The simulation model can be imported in any virtualization application. The following guide shows the procedure in Oracle VM VirtualBox, which is an open-source and is available for all major operating systems including Windows, Linux, OS X, and Solaris.

The tool can be downloaded on the following webpage: https://www.virtualbox.org/

Import in the VirtualBox can be done by clicking on “File / Import Appliance” and then choosing the ova file. This action will create a new virtual machine called “SmartGridSim”, which should appear in the list of virtual machines in the left part of the window as illustrated in Figure 4.

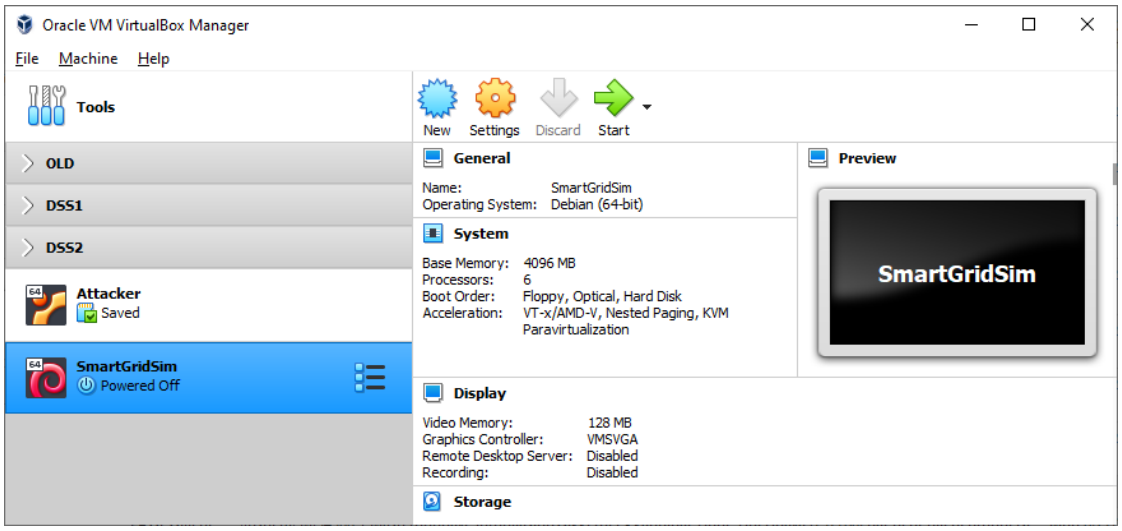

Figure 4 - Imported virtual machine of the simulation model

Virtual machine configuration

The virtual machine can be configured by clicking on the *SmartGridSim* machine and clicking on Machine / Settings. This will open the configuration window with several sections. The most important ones are described below:

- System – contains performance configuration including RAM (base memory) and processor (number of processors and execution cap). The simulation model should be configured with at least 4 GB of RAM and 2 processors with 100% execution cap.
- Display – contains video memory configuration and configuration of multiple monitors.
- Network – contains virtual network interface cards, which can be used to connect another (virtual) machine into the model, or to connect the virtual machine to the internet.
- Shared Folders – contains configuration of shared folders, which can be used to exchange files between the host OS and the virtual machine.
- User Interface – contains options for displaying control elements during the VM run.

Source file structure

The simulation model source files are located in the folder */home/sgsim/SmartGridSim*

The folder contains the following subfolders:

- comlib_dps – contains library files for digital primary substation communication emulation (GOOSE, SV).
- comlib_dss – contains library files for digital secondary substation communication emulation (IEC104).
- logs – stores log files from performance measurements.
- topologies – contains topology scripts for the following models:
    - DSSTopology.py – topology contains two digital secondary substations and the control center.
    - SmartGridTopology.py – extended model which adds one primary substation to the topology.

Scripts for starting the models are placed in the *SmartGridSim* folder - *StartDSSTopology.sh* and *StartSGTopology.sh*. These scripts can be launched from the Terminal with the following commands[3] (require sudo privileges):

```
./StartDSSTopology.sh
./StartSGTopology.sh
```

Start of the model

Start of the virtual machine

The virtual machine with the model can be started by double-clicking on the *SmartGridSim* virtual machine in the list, or by selecting it and pressing the green arrow button *Start*.

A window with the virtual machine should appear and it should boot the guest operating system and display a login screen. The following credentials are used for login:

- Username: sgsim
- Password: sgsim

User interface

The virtual machine uses default user interface of the Linux Lite - Xfce. This is a lightweight, simple, and easy to use interface, which uses layout similar to Windows operating system. The main screen is shown in Figure 5.

[3] Executed from the folder /home/sgsim/SmartGridSim/

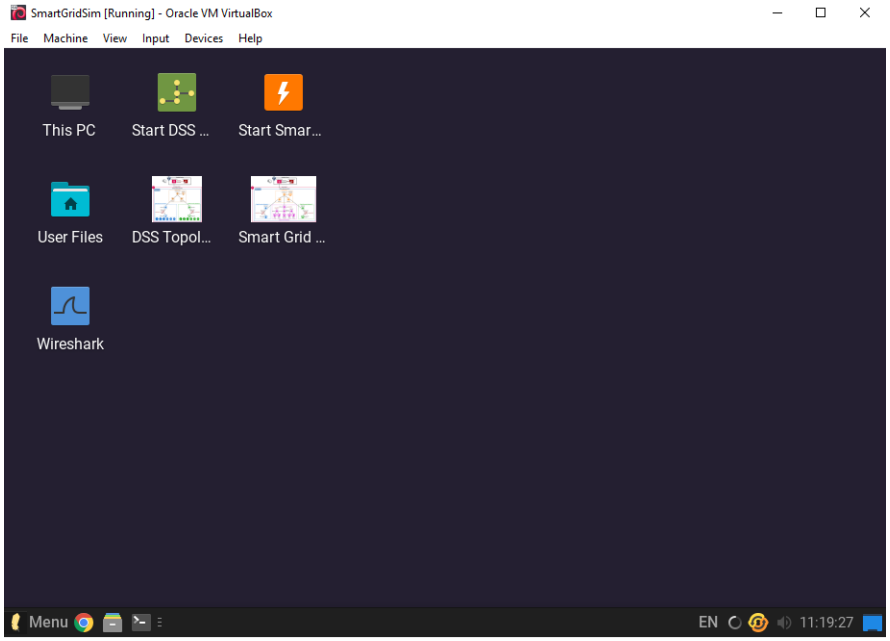

Figure 5 – User interface of the virtual machine

Start of the model

The model can be started by clicking on the icon *Start DSS Model* (for the basic topology) or *Start Smart Grid Model* (extended topology) located on the desktop. This file executes launch scripts located in */home/sgsim/SmartGridSim/*

Alternatively, the simulation model can be launched from the terminal[4] using the following commands (it will require typing the login password to get the superuser privileges):

```
sudo python2 ~/SmartGridSim/topologies/DSSTopology.py
sudo python2 ~/ SmartGridSim /topologies/SmartGridTopology.py
```

The scripts will start the Mininet network emulator with the appropriate topology. Topologies figures are placed on the desktop in form of pictures.

Start of the model takes a few seconds during which an output is displayed in the terminal window. It informs about each phase such as "*Creating hosts*", "*Adding links*", or "*Starting switches*". The model is running when the terminal line[5] displays the keyword *mininet>* as shown in Figure 6.

[4] Terminal shortcut is placed on the bottom panel, or it can be launched from the Menu and typing "Terminal Emulator".
[5] The simulation model is running in the Mininet emulator and it uses its components.

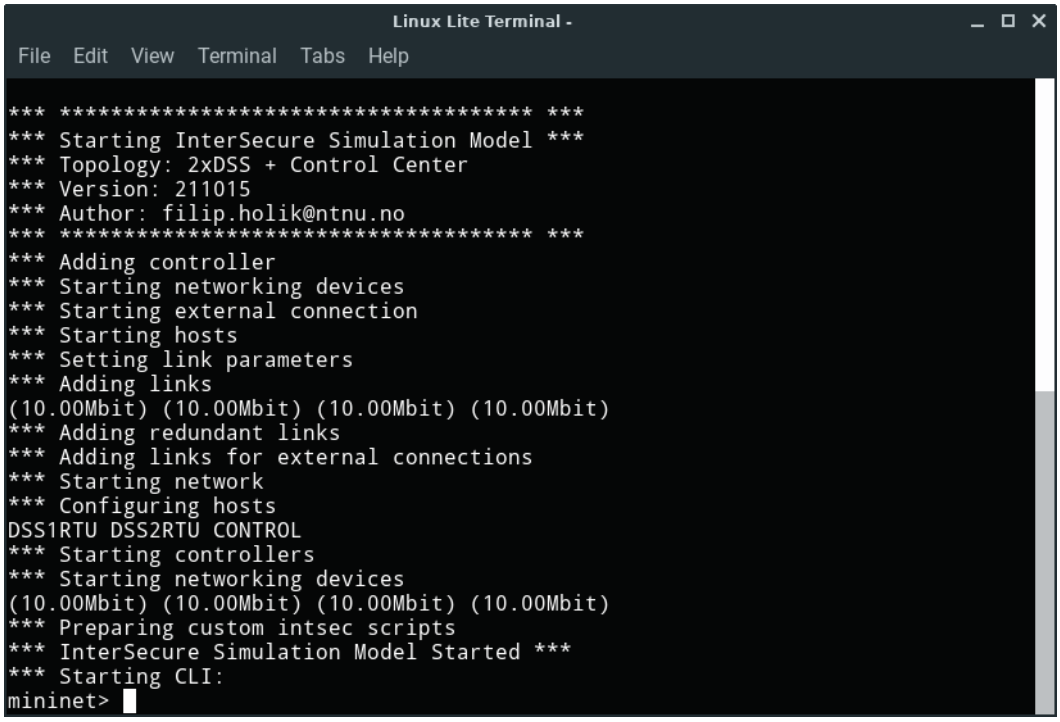

Figure 6 – Running simulation model topology in the Mininet network emulator

Working with the model

The basic control

All the available commands of the Mininet can be listed with the *help* command:

```
mininet> help
```

Help with a specific command can be displayed as follows (example shows *intsec_startcom* command):

```
mininet> help sgsim_startcom_104
Starts the IEC104 communication (periodical and read requests) for both secondary substations.
```

Custom commands developed specifically for this project start with a word *sgsim* and they include the following:

- sgsim_attackmirror – makes DSS ASW devices to mirror traffic to external connections. This can be used for traffic sniffing or man-in-the-middle type of attacks.
- sgsim_startcom_104 – starts the IEC104 communication (periodical and read requests) for both secondary substations.
- sgsim_startperfmon – starts the IEC104 communication (periodical and read requests) with performance monitoring.

The IEC104 communication

The IEC104 communication can be started by typing the *sgsim_startcom_104* command:

```
mininet> sgsim_startcom_104
```

The command will execute scripts, which will launch communication between RTUs and control center. The command will open four terminal windows with the following content:

- DSS1RTU – sends periodical messages from DSS1 and replies for read requests from the control device.
- DSS2RTU – sends periodical messages from DSS2 and replies for read requests from the control device.
- Control – DSS1 Monitoring – displays received messages from DSS1.
- Control – DSS2 Monitoring – displays received messages from DSS2.

Running IEC104 communication is shown in Figure 7.

The communication can be stopped by closing all four opened windows (alternatively by stopping the scripts with the *ctrl+c* shortcut).

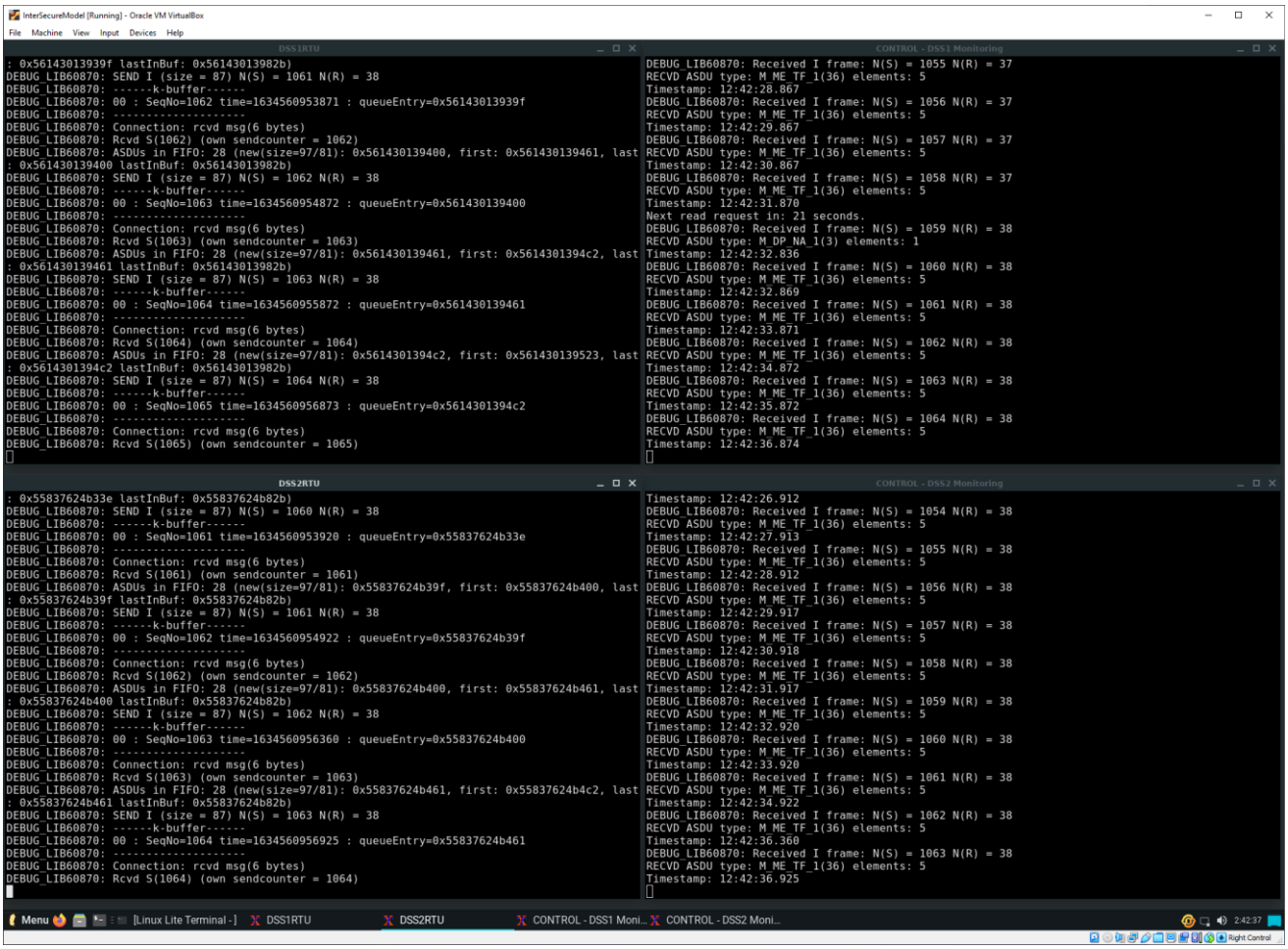

Figure 7 – Running IEC104 communication

Other communication

The extended topology supports additional two types of communication:

- *sgsim_startcom_goose* – starts GOOSE communication between IED1, IED4 and DPS HMI.
- *sgsim_startcom_sv* – starts SV communication between IED2, IED3 and DPS HMI.

Traffic analysis

The virtual machine includes Wireshark network analyzer which can be used for displaying the model communication. Wireshark can be launched from *Menu/Internet/Network Analyzer*, or from the desktop shortcut.

Wireshark is configured to run with root privileges and it automatically re-scan for interfaces upon the start / stop of the model, so it can be launched in any order.

Performance measurement

The communication can be started with automated performance measurement. This is conducted when the command *sgsim_startperfmon* is executed. The command will open the same four windows, but the *Control* windows will display performance parameters connected with every message. They include:

- Delay values – current, maximum, average, and number / percentage of packets above set threshold.
- IPDV (Inter-Packet Delay Variation) values – current, minimum, maximum, average, and number / percentage of packets above set threshold.
- Jitter values – current, maximum and average.
- Lost packets – number in percentage is displayed only when the packets are no longer transmitted.

The results are automatically logged into a file located in the folder */home/sgsim/SmartGridSim/logs/*. The logged files have the following filename format: *dss_X_perf_log_YYMMDD-HHMMSS.MSS.csv*[6]

Configuring performance thresholds

The automated performance measurement has defined thresholds for maximum delay, maximum IPDV and maximum acceptable packet loss. Measurements exceeded configured delay and IPDV values are counted and if their percentage is higher than acceptable threshold (or packet loss is above threshold), the measurement will be classified as failed. Otherwise, it will be passed. All the variables are defined in the file: */home/sgsim/SmartGridSim/comlib_dss/sgdevices/PERFMON/perfmon.c*

The following list describes the configuration options:

- MAX_DELAY_MS – uses a difference between IEC104 message timestamp and time of packet arrival. The default value is set to 500 ms.

[6] X identifies digital secondary substation (1 or 2), YYMMDD states date and HHMMSS.MSS states time to millisecond precision.

- MAX_IPDV – uses a difference between times of two consequent packets arrival times minus messages sending frequency (1 second). The default value is set to 500 ms.
- MAX_PACKET_LOSS – is calculated in percentage as a number of received packets and a number of expected packets (calculated as sending frequency * length of the experiment). The default value is set to 1%.
- MAX_ACCEPTABLE_EXCEED_PERCENT – defines the maximum acceptable amount of delay and IPDV packets above the set threshold. The default value is set to 1%.
- EXPERIMENT_LENGTH_SEC – defines the length of the measurement. The default value is set to 60 seconds.

Connecting an external machine

The simulation model can be connected to an external machine, which can be either physical, or virtual. This can be used for example for connecting an attacker with Kali Linux to perform testing attacks. The following guide describes connecting a Kali Linux virtual machine in the Oracle VM VirtualBox.

External virtual machine configuration

The external virtual machine has to be configured with the following network adapter (Settings/Network/Adapter X[7]):

- Network Adapter enabled
- Attached to: Internal network
- Name: AttackDSS1 / AttackDSS2 / AttackDPS1 / AttackDPS2*
- Promiscuous mode: Allow VMs

* This name must be the same on the adapter of the model.

The appropriate interface must be configured inside the virtual machine. The example configuration is:

- IP Address: 1.1.1.100
- Netmask: 8
- Gateway: 1.1.1.1

Note: In case of a simulated attack, the attacker would have to first get the addressing information by packet sniffing.

Illustration of the required configuration is shown in Figure 8.

[7] Can be any of the interfaces (1 – 4)

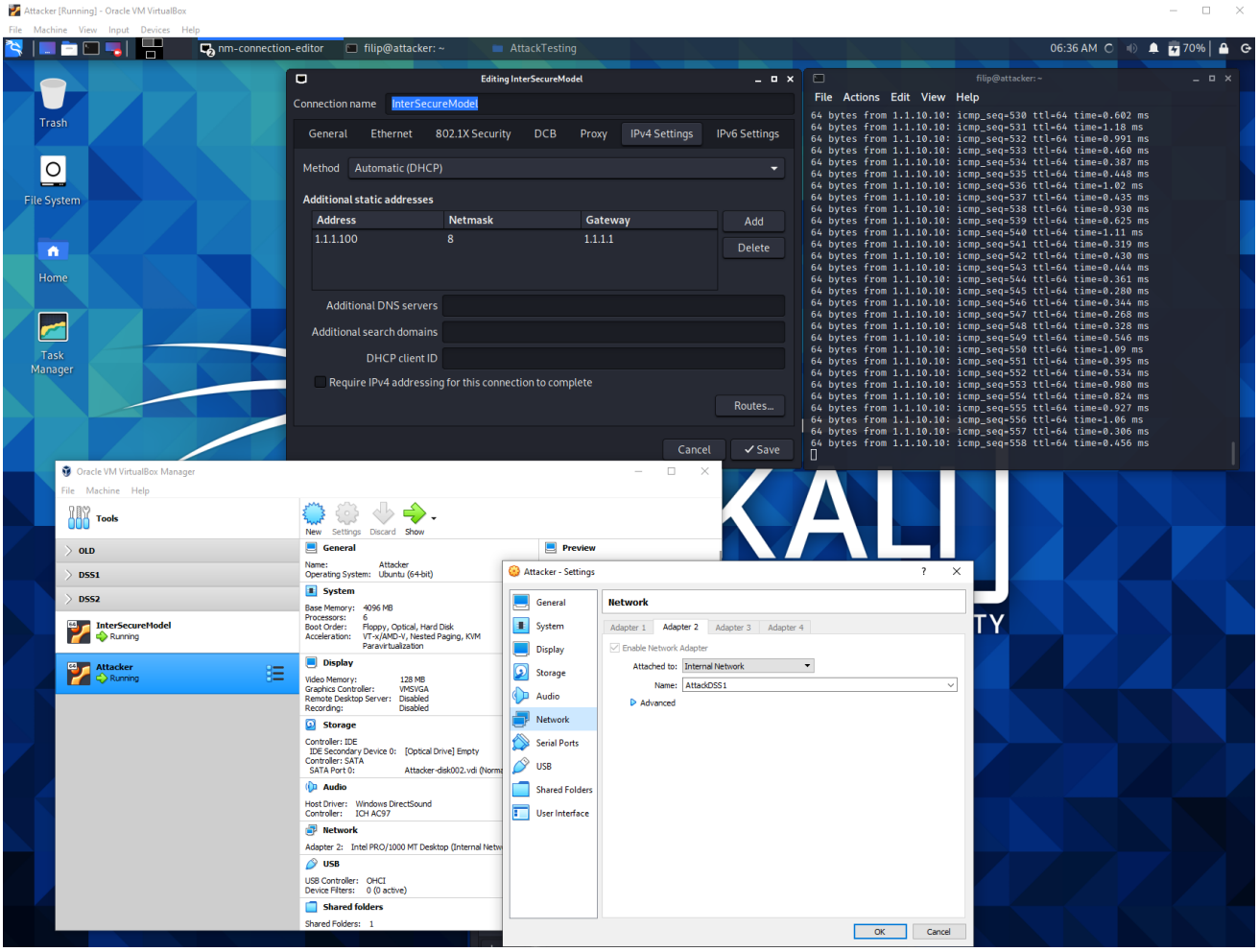

Figure 8 – Configuration of an external virtual machine

SCADA Simulation

Introduction

The simulation model contains a simple SCADA simulation feature, which uses a PHP server, SQL database and HTML for visual representation. It monitors status of the smart grid and displays received data from substation devices. It uses the *extended topology* for visualization[8].

Functionality

The SCADA simulation has the following functionality divided in three web pages:

- SCADA Simulation
  - Provides two buttons to navigate between the following pages.
- Quick overview
  - Shows status of IEC104 communication from connected secondary substations. It displays status (connected, disconnected), timestamp of the last received IEC104 message and its ASDU type.
- Topology
  - Shows the smart grid topology as shown in Figure 9. It visualizes smart grid status and displays last received data from various devices.

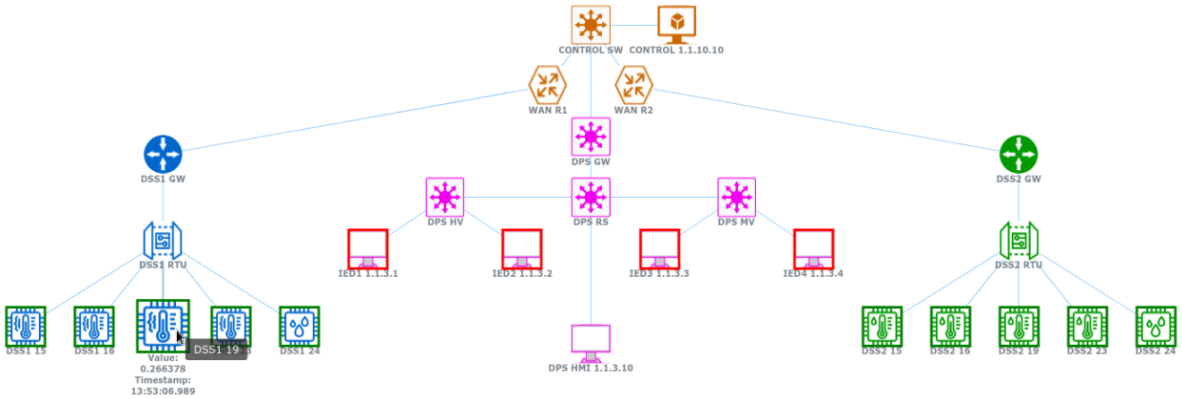

Figure 9 – Topology page with two DSS connected

Start of the SCADA simulation

The SCADA simulation server can be started from the desktop by clicking on the "Start Server" icon and "SCADA GUI" icon. It can also be launched manually from the *comlib_dss/sgdevices/PHPserver* folder with the command:

```
php -S localhost:8000
```

[8] It can also be used for the *basic topology.* In that case, the primary substation will be indicated as not connected (red color).

Quick overview

The quick overview page shows the state of the DSSs, the timestamp and the type of the last received ASDU message as shown in Figure 10. It does not display primary substation communication as these messages (GOOSE, SV) are not routed outside the substation.

DSS	State	Timestamp	RCVD ASDU type
1	Up	12:12:55.157	M_ME_TF_1(36)
2	Up	12:12:55.554	M_ME_TF_1(36)

Figure 10 – Quick overview page with two DSS connected

Topology

The topology page shows an overview of the entire smart grid network (extended topology) with additional information. The basic overview is shown in Figure 9 (DSS connected) and Figure 11 (all substations disconnected).

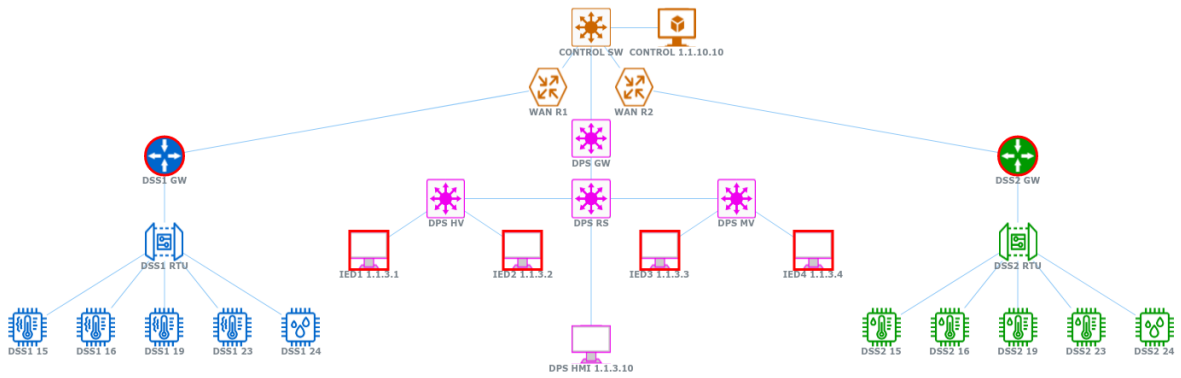

Figure 11 – Topology page with all substations disconnected

Data from each device can be displayed by hovering the cursor over the device. This will show data from the last receive message. The example on Figure 12 shows data from DSS1 sensor 24 (value and timestamp) on the left side and data from a substation IED on the right side.

Figure 12 – Topology with IED2 connected

Integrated attacks

The extended topology contains two integrated attacks – denial of service and false data injection attacks.

Denial of service attack

The following script will “compromise” DSS1RTU device which will start flooding the control center with a lot of traffic:

- *sgsim_attack_dos* – the script will start the hping3 tool in the SYN flooding mode, generating as much traffic as possible. The same attack can be launched from any device (xterm) with the command: *sudo hping3 -S—flood 1.1.10.10.*

False data injection attack

The extended topology contains an attacker device (named ATTACKER) connected to the DPS RS. The topology contains the following prepared script which can automatically perform the “*false data injection*” attack:

- *sgsim_attack_goose_fdi* – the attack sends modified GOOSE messages with floating-point data value changed from the 5 range to 10 range, simulating a value, which would result to circuit breaker trigger (for example too high oil temperature, too high phase, etc.). The attack can be verified in the SCADA simulation as shown in Figure 13.

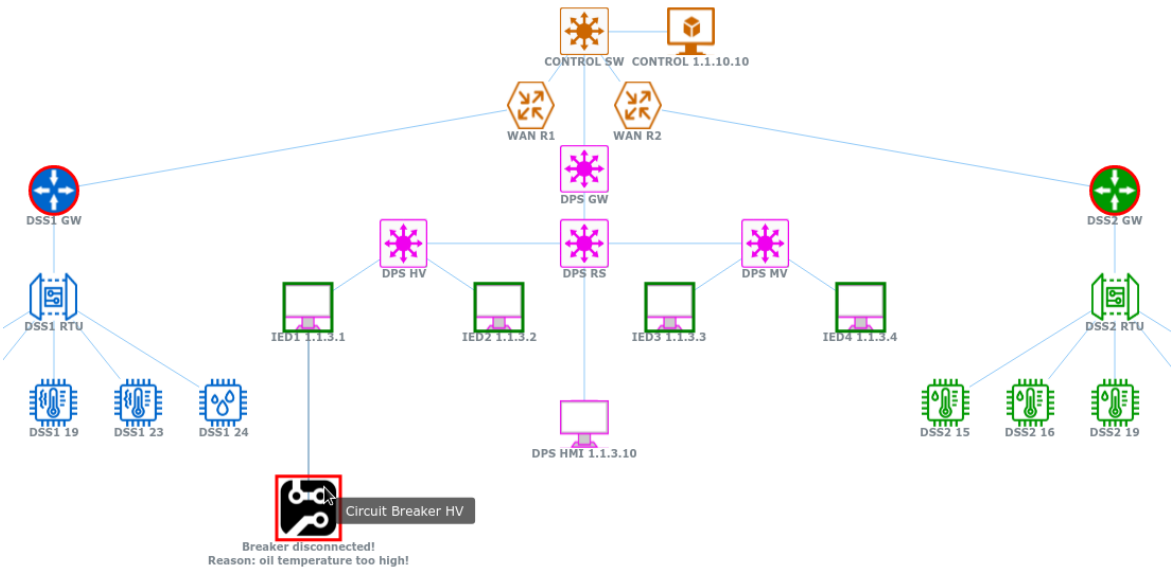

Figure 13 – False data injection attack triggering the circuit breaker

DoS detection

The SCADA simulation has a feature to detect a DoS (Denial of Service) attack and visualize it in the topology page, where the problematic substation can be disconnected. This detection can be activated by launching the *DDOSDetection.sh* script with the command:

```
cd comblib_dss/sgdevices
sudo ./DDOSDetection.sh
```

This detection is automatically started if the SCADA simulation is launched from the desktop icon "*Start Server*".

The detection script listens on the control center port and if it detects an overflow of messages (by default 100 packets per second, but can be configured in the script), it will indicate it in the topology as shown in Figure 14. The script opens a new terminal window where current numbers of packets per seconds from connected secondary substations are shown – Figure 15.

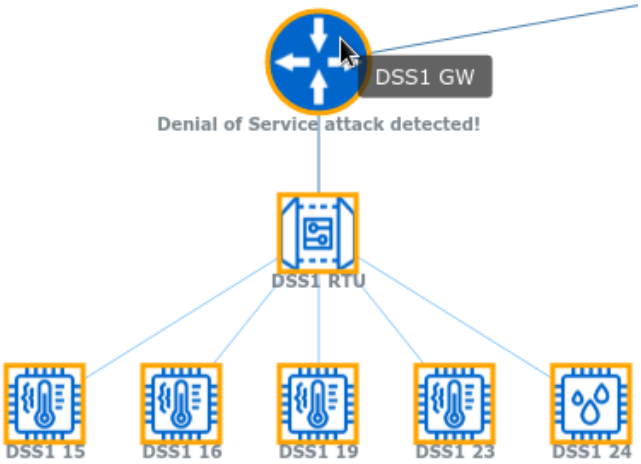

Figure 14 – DoS detected on data coming from DSS1

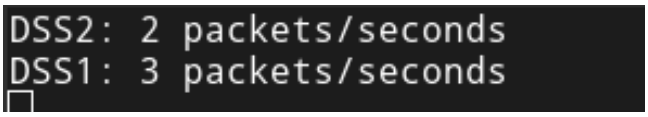

Figure 15 – Number of packets per seconds from each DSS

DSS connection control

Any substation can be disconnected / connected from the control center (for example in case of a detected DoS attack) by clicking the button in the topology page. Figure 16 shows a situation with DSS1 disconnected and DSS2 connected with a DoS attack in progress.

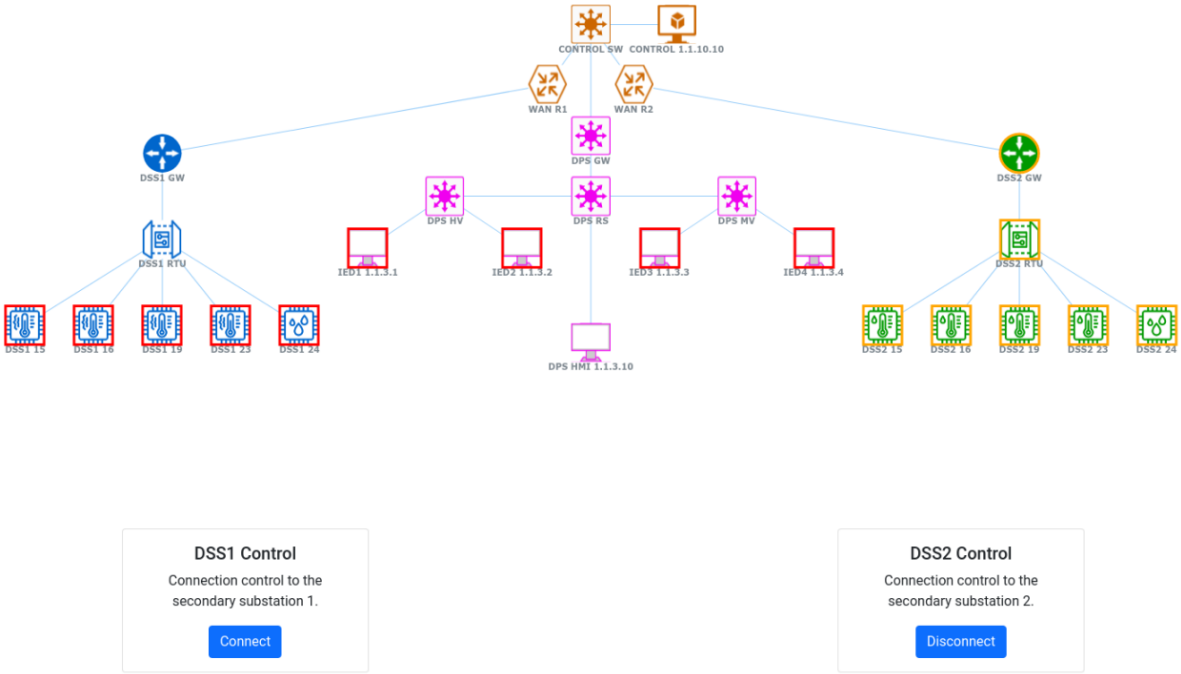

Figure 16 – Network under DDOS attack from DDS1 and with connect and disconnect button